2.35 Volume of Revolution – Part 2

Worked Solutions

1a)

$$V = \pi \int_1^{14} y^2 \, dx = \pi \int_1^{14} (2x-1)^{2/3} \, dx = \frac{3\pi}{10} \left[(2x-1)^{5/3} \right]_1^{14} = \frac{363\pi}{5} \text{ units}^3.$$

1b)

$$V = \pi \int_{\pi/6}^{\pi/3} \sec^2 x \, dx = \pi [\tan x]_{\pi/6}^{\pi/3} = \pi \left(\sqrt{3} - \frac{1}{\sqrt{3}} \right) = \frac{2\sqrt{3}\pi}{3} \text{ units}^3.$$

1c)

$$\begin{aligned} V &= \pi \int_1^2 \ln x \, dx = \pi [x \ln x - x]_1^2 \\ &= \pi (2 \ln 2 - 1) = \pi (\ln 4 - 1) \text{ units}^3. \end{aligned}$$

1d)

$$V = \pi \int_0^{\pi/2} \frac{2 \cos x}{1 + \sin x} \, dx = 2\pi [\ln(1 + \sin x)]_0^{\pi/2} = \pi \ln 4 \text{ units}^3.$$

1e)

By integration by parts,

$$\int x \operatorname{cosec}^2 x \, dx = -x \cot x + \ln |\sin x| + c.$$

Therefore

$$\begin{aligned} V &= \pi [-x \cot x + \ln |\sin x|]_{\pi/4}^{\pi/2} \\ &= \pi \left(\frac{\pi}{4} + \frac{1}{2} \ln 2 \right) \text{ units}^3. \end{aligned}$$

1f)

$$\begin{aligned} V &= \pi \int_0^{\pi/4} (\sec x + \tan x)^2 \, dx \\ &= \pi \int_0^{\pi/4} (2 \sec^2 x + 2 \sec x \tan x - 1) \, dx \\ &= \pi [2 \tan x + 2 \sec x - x]_0^{\pi/4} = \pi \left(2\sqrt{2} - \frac{\pi}{4} \right) \text{ units}^3. \end{aligned}$$

2a)

The curve meets the x-axis when $\sqrt{\cos x} = 0$, so $x = \pi/2$. Hence

$$V = \pi \int_{\pi/4}^{\pi/2} \cos x \, dx = \pi \left(1 - \frac{\sqrt{2}}{2}\right) \text{ units}^3.$$

2b)

The curve meets the x-axis when $\sqrt{x} \ln x = 0$, so $x = 1$ in the relevant region. By integration by parts,

$$\int x(\ln x)^2 \, dx = \frac{x^2}{2}(\ln x)^2 - \frac{x^2}{2} \ln x + \frac{x^2}{4} + c.$$

Thus

$$V = \pi \int_1^4 x(\ln x)^2 \, dx = \pi \left(8(\ln 4)^2 - 8 \ln 4 + \frac{15}{4}\right) \text{ units}^3.$$

3a)

The region is the area under $y = \sin x$ and above the x-axis for $0 \leq x \leq \pi$.

3b)

$$V = \pi \int_0^{\pi} \sin^2 x \, dx = \frac{\pi}{2} \int_0^{\pi} (1 - \cos 2x) \, dx = \frac{\pi^2}{2} \text{ units}^3.$$

4a)

The region is under $y = \sinh x$ and above the x-axis for $0 \leq x \leq \ln 5$.

4b)

Using $\sinh^2 x = \frac{1}{2}(\cosh 2x - 1)$,

$$\begin{aligned} V &= \pi \int_0^{\ln 5} \sinh^2 x \, dx = \pi \left[\frac{1}{4} \sinh 2x - \frac{x}{2}\right]_0^{\ln 5} \\ &= \pi \left(\frac{78}{25} - \frac{1}{2} \ln 5\right) \text{ units}^3. \end{aligned}$$

5a)

The region is under $y = \cosh x$ and above the x-axis for $0 \leq x \leq \ln 3$.

5b)

Using $\cosh^2 x = \frac{1}{2}(\cosh 2x + 1)$,

$$\begin{aligned} V &= \pi \int_0^{\ln 3} \cosh^2 x \, dx = \pi \left[\frac{1}{4} \sinh 2x + \frac{x}{2}\right]_0^{\ln 3} \\ &= \pi \left(\frac{10}{9} + \frac{1}{2} \ln 3\right) \text{ units}^3. \end{aligned}$$

6a)

The non-zero x-intercept satisfies $2x\cos x = 0$, giving $x = \pi/2$. Hence $P = \pi/2$.

6b)

Since $y^2 = 2x\cos x$,

$$\begin{aligned} V &= \pi \int_0^{\pi/2} 2x\cos x\,dx \\ &= \pi[2x\sin x + 2\cos x]_0^{\pi/2} = (\pi^2 - 2\pi) \text{ units}^3. \end{aligned}$$

7a)

At the midpoint, the radius is half of the 2 cm diameter, so $a = 1$.

7b)

For Model A, $2.5 = 1 + 36k$, so $k = 1/24$. For Model B, $2.5 = 1 + m(1 - \cos\pi)$, so $m = 3/4$.

7c)

For Model A, $y = 1 + x^2/24$ and

$$\begin{aligned} V_A &= \pi \int_{-6}^{6} \left(1 + \frac{x^2}{24}\right)^2 dx \\ &= \pi \left[x + \frac{x^3}{36} + \frac{x^5}{2880}\right]_{-6}^{6} = \frac{147\pi}{5} \approx 92.4 \text{ cm}^3. \end{aligned}$$

For Model B, $y = \frac{7}{4} - \frac{3}{4}\cos(\pi x/6)$. Squaring, using $\cos^2 A = \frac{1}{2}(1 + \cos 2A)$, and integrating from -6 to 6 gives

$$V_B = \frac{321\pi}{8} \approx 126 \text{ cm}^3.$$

7d)

Model A is quadratic, so its gradient increases linearly away from the midpoint and the curve becomes steeper towards the ends. Model B has zero gradient at both the midpoint and the ends, so it flattens near the ends. Model A better fits the described change.

8a)

Let

$$\frac{6x + 11}{(x + 3)(2x - 1)} = \frac{A}{x + 3} + \frac{B}{2x - 1}.$$

Then $6x + 11 = A(2x - 1) + B(x + 3)$. Taking $x = -3$ gives $A = 1$; taking $x = 1/2$ gives $B = 4$. Therefore

$$\frac{6x + 11}{(x + 3)(2x - 1)} = \frac{1}{x + 3} + \frac{4}{2x - 1}.$$

8b)

$$\begin{aligned} V &= \pi \int_1^4 \frac{6x + 11}{(x + 3)(2x - 1)}\,dx \\ &= \pi[\ln|x + 3| + 2\ln|2x - 1|]_1^4 = \pi \ln\left(\frac{343}{4}\right) \text{ units}^3. \end{aligned}$$

9)

$$\frac{(3x+4)^2}{(x+2)^2x} = \frac{9x^2+24x+16}{x(x+2)^2} = \frac{4}{x} + \frac{5}{x+2} - \frac{2}{(x+2)^2}.$$

Therefore

$$\begin{aligned} V &= \pi \int_1^3 \left(\frac{4}{x} + \frac{5}{x+2} - \frac{2}{(x+2)^2} \right) dx \\ &= \pi \left[4\ln|x| + 5\ln|x+2| + \frac{2}{x+2} \right]_1^3 \\ &= \pi \left(5\ln 5 - \ln 3 - \frac{4}{15} \right) = \pi \left(\ln \frac{5^5}{3} - \frac{4}{15} \right) \text{ units}^3. \end{aligned}$$

Thus $a = 5$ and $b = 1$.

10a)

The volume is

$$V = \pi \int_0^{k/2} \frac{dx}{k - x^2}.$$

Using partial fractions,

$$\frac{1}{k-x^2} = \frac{1}{2\sqrt{k}} \left(\frac{1}{\sqrt{k}+x} + \frac{1}{\sqrt{k}-x} \right).$$

Hence

$$V = \frac{\pi}{2\sqrt{k}} \left[\ln \left(\frac{\sqrt{k}+x}{\sqrt{k}-x} \right) \right]_0^{k/2} = \frac{\pi}{2\sqrt{k}} \ln \left(\frac{2\sqrt{k}+k}{2\sqrt{k}-k} \right).$$

10b)

The condition gives $k/2 = 1/2$, so $k = 1$. Therefore

$$V = \frac{\pi}{2} \ln 3 \text{ units}^3.$$

11a)

The two curves meet at $x = 5\pi/4$; sketch the region between them from $x = \pi/4$ to $x = 5\pi/4$.

11b)

The difference of the squares of the radii is

$$(1+\sin x)^2 - (1+\cos x)^2 = 2\sin x - 2\cos x - \cos 2x.$$

Therefore

$$\begin{aligned} V &= \pi \int_{\pi/4}^{5\pi/4} ((1+\sin x)^2 - (1+\cos x)^2) dx \\ &= \pi [-2\cos x - 2\sin x - \tfrac{1}{2}\sin 2x]_{\pi/4}^{5\pi/4} = 4\sqrt{2}\pi. \end{aligned}$$

12a)

From $z - z^{-1} = 2i\sin\theta$,

$$\begin{aligned}(2i\sin\theta)^4 &= (z - z^{-1})^4 \\ &= z^4 - 4z^2 + 6 - 4z^{-2} + z^{-4} \\ &= 2\cos 4\theta - 8\cos 2\theta + 6.\end{aligned}$$

Thus $\sin^4\theta = \frac{1}{8}\cos 4\theta - \frac{1}{2}\cos 2\theta + \frac{3}{8}$, so $a = 1/8$, $b = -1/2$ and $c = 3/8$.

12b)

Using $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ and part (a),

$$(1 + \sin^2 x)^2 = \sin^4 x + 2\sin^2 x + 1 = \frac{1}{8}\cos 4x - \frac{3}{2}\cos 2x + \frac{19}{8}.$$

Hence

$$V = \pi\int_0^\pi (1 + \sin^2 x)^2 dx = \frac{19\pi^2}{8} \text{ units}^3.$$

13a)

Let $u = 2 - \cos x$, so $du = \sin x\, dx$. When $x = 0$, $u = 1$, and when $x = \pi$, $u = 3$. Therefore

$$\int_0^\pi \frac{\sin x}{\sqrt{2 - \cos x}} dx = \int_1^3 u^{-1/2} du = 2\sqrt{3} - 2.$$

13b)

Since $y^2 = 4\sin x(2 - \cos x)^{-1/2}$,

$$V = 4\pi\int_0^\pi \frac{\sin x}{\sqrt{2 - \cos x}} dx = 8\pi(\sqrt{3} - 1) \text{ units}^3.$$

14)

$$\begin{aligned}V &= \pi\int_0^{\ln k} xe^{2x}\, dx = \pi\left[\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x}\right]_0^{\ln k} \\ &= \pi\left(\frac{1}{2}k^2\ln k - \frac{1}{4}k^2 + \frac{1}{4}\right).\end{aligned}$$

Equating with $\pi(81\ln 3 - 20)$ gives $k = 9$.

15)

Write $2\cos x + \sqrt{5}\sin x = 3\cos(x - \alpha)$, where $\cos\alpha = 2/3$ and $\sin\alpha = \sqrt{5}/3$. Then

$$\begin{aligned}V &= \pi\int_0^{\pi/4} \left(\frac{8}{3\cos(x - \alpha)}\right)^2 dx \\ &= \frac{64\pi}{9}[\tan(x - \alpha)]_0^{\pi/4} \\ &= 32\pi(\sqrt{5} - 2) \text{ units}^3.\end{aligned}$$

16a)

Using $z - z^{-1} = 2i\sin\theta$ and $z^n + z^{-n} = 2\cos n\theta$,

$$\begin{aligned}(2i\sin\theta)^6 &= (z - z^{-1})^6 \\ &= 2\cos 6\theta - 12\cos 4\theta + 30\cos 2\theta - 20.\end{aligned}$$

Since $(2i)^6 = -64$, division by -64 gives

$$\sin^6\theta = \frac{5}{16} - \frac{15}{32}\cos 2\theta + \frac{3}{16}\cos 4\theta - \frac{1}{32}\cos 6\theta.$$

16b)

At $x = 4$, the sine term is $\sin(\pi/2) = 1$, so the maximum height is H. Thus $H = 3$.

16c)

$$\begin{aligned}V &= 9\pi\int_0^4 \sin^6\left(\frac{\pi x}{8}\right)dx \\ &= 9\pi\int_0^4 \left(\frac{5}{16} - \frac{15}{32}\cos\frac{\pi x}{4} + \frac{3}{16}\cos\frac{\pi x}{2} - \frac{1}{32}\cos\frac{3\pi x}{4}\right)dx \\ &= \frac{45\pi}{4}\ \text{cm}^3.\end{aligned}$$

16d)

The model assumes the ornament is a perfectly smooth solid of revolution and does not account for the thickness of its glass walls.

17)

The intersections solve $x = 1/(2\sqrt{2} - x)$, so

$$x(2\sqrt{2} - x) = 1 \quad\Longrightarrow\quad x = \sqrt{2} - 1 \text{ or } x = \sqrt{2} + 1.$$

The region lies between these two values. Since $y^2 = x^2$ on the line and $y^2 = (2\sqrt{2} - x)^{-2}$ on the curve,

$$\begin{aligned}V &= \pi\int_{\sqrt{2}-1}^{\sqrt{2}+1}\left(x^2 - \frac{1}{(2\sqrt{2} - x)^2}\right)dx \\ &= \pi\left[\frac{x^3}{3} - \frac{1}{2\sqrt{2} - x}\right]_{\sqrt{2}-1}^{\sqrt{2}+1} = \frac{8\pi}{3}\ \text{units}^3.\end{aligned}$$

18a)

From $r = 3/(2 - \cos\theta)$, $2r - r\cos\theta = 3$. Since $r\cos\theta = x$, $r = (x + 3)/2$. Substituting $r^2 = x^2 + y^2$ gives

$$y^2 = \frac{(x+3)^2}{4} - x^2.$$

18b)

The equation contains y only as y^2. Therefore replacing y with $-y$ leaves the equation unchanged, so the curve is symmetrical about the x-axis.

18c)

The positive x-intercept is $x = 3$. Hence

$$\begin{aligned} V &= \pi \int_0^3 \left(\frac{(x+3)^2}{4} - x^2 \right) dx \\ &= \pi \left[-\frac{x^3}{4} + \frac{3x^2}{4} + \frac{9x}{4} \right]_0^3 = \frac{27\pi}{4} \text{ units}^3. \end{aligned}$$

19a)

From $r = 3 - 2\sec\theta$, multiply by $\cos\theta$ and use $r\cos\theta = x$:

$$xr = 3\cos\theta\, r - 2r = 3x - 2r.$$

Thus $r(x+2) = 3x$, so $r = 3x/(x+2)$. Substituting $r^2 = x^2 + y^2$ gives

$$y^2 = \frac{9x^2}{(x+2)^2} - x^2.$$

19b)

The Cartesian equation contains y only as y^2. Replacing y by $-y$ leaves the equation unchanged, so the curve is symmetrical about the x-axis.

19c)

$$\begin{aligned} y^2 &= \frac{9x^2}{(x+2)^2} - x^2 \\ &= 9 + \frac{-36x - 36}{(x+2)^2} - x^2 \\ &= 9 + \frac{36}{(x+2)^2} - \frac{36}{x+2} - x^2. \end{aligned}$$

19d)

$$\begin{aligned} V &= \pi \int_0^1 \left(9 + \frac{36}{(x+2)^2} - \frac{36}{x+2} - x^2 \right) dx \\ &= \pi \left[9x - \frac{36}{x+2} - 36\ln|x+2| - \frac{x^3}{3} \right]_0^1 \\ &= \pi \left(\frac{44}{3} + 36\ln\frac{2}{3} \right) \text{ units}^3. \end{aligned}$$

20a)

$$V = \pi \int_0^{1/2} \frac{dx}{\sqrt{1-x^2}} = \pi \left[\arcsin x\right]_0^{1/2} = \frac{\pi^2}{6} \text{ units}^3.$$

20b)

$$V = 9\pi \int_0^4 \frac{dx}{x^2+9} = 3\pi \left[\arctan\left(\frac{x}{3}\right)\right]_0^4 = 3\pi \arctan\frac{4}{3} \text{ units}^3.$$

20c)

Completing the square gives $8 - 4x - 4x^2 = 9 - 4(x + \frac{1}{2})^2$. Thus

$$\begin{aligned} V &= \pi \int_0^{1/4} \frac{dx}{\sqrt{8-4x-4x^2}} \\ &= \frac{\pi}{2} \left[\arcsin\left(\frac{2x+1}{3}\right)\right]_0^{1/4} \\ &= \frac{\pi}{12}\left(\pi - 6\arcsin\frac{1}{3}\right) \text{ units}^3. \end{aligned}$$

20d)

$$V = \pi \int_0^{\sqrt{6}} \frac{dx}{x^2+2} = \frac{\pi}{\sqrt{2}} \left[\arctan\left(\frac{x}{\sqrt{2}}\right)\right]_0^{\sqrt{6}} = \frac{\pi^2}{3\sqrt{2}} \text{ units}^3.$$

20e)

Using $\int dx/(x^2+9)^2 = x/(18(x^2+9)) + \arctan(x/3)/54$,

$$V = \pi \int_0^1 \frac{dx}{(x^2+9)^2} = \frac{\pi}{54}\left(\arctan\frac{1}{3} + \frac{3}{10}\right) \text{ units}^3.$$

21)

Using partial fractions,

$$\frac{1}{4-x^2} = \frac{1}{4}\left(\frac{1}{2-x} + \frac{1}{2+x}\right).$$

Therefore

$$\begin{aligned} V &= \pi \int_0^1 \frac{dx}{4-x^2} = \frac{\pi}{4}\left[-\ln|2-x| + \ln|2+x|\right]_0^1 \\ &= \frac{\pi}{4}\ln 3 \text{ units}^3. \end{aligned}$$

Alternatively, $V = \pi[\operatorname{artanh}(x/2)]_0^1 = \frac{\pi}{4}\ln 3$.

22)

The volume is

$$V = \pi \int_0^2 \frac{dx}{x^2+3}.$$

Using $\int dx/\sqrt{x^2+a^2} = \operatorname{arsinh}(x/a)$,

$$\begin{aligned} V &= \pi \int_0^2 \frac{dx}{\sqrt{x^2+3}} = \pi \left[\operatorname{arsinh}\left(\frac{x}{\sqrt{3}}\right)\right]_0^2 \\ &= \pi \ln\left(\frac{2+\sqrt{7}}{\sqrt{3}}\right) \text{ units}^3. \end{aligned}$$

23a)

For Model A, fitting $y = a + bx$ to the data gives $a = -1.2$ and $b = 2$.

23b)

For Model B, fitting $y = p/(x^2+q)$ gives $p = 2$ and $q = 1$.

23c)

For Model A, $y = 0.2x^2 - 1.2x + 2$. Therefore

$$\begin{aligned} V &= \pi \int_0^2 (0.2x^2 - 1.2x + 2)^2 \, dx \\ &= \pi \left[4x - 2.4x^2 + \frac{56}{75}x^3 - 0.12x^4 + 0.008x^5\right]_0^2 \\ &\approx 8.51 \text{ m}^3. \end{aligned}$$

23d)

For Model B,

$$\begin{aligned} V &= \pi \int_0^2 \left(\frac{2}{x^2+1}\right)^2 dx = 4\pi \int_0^2 \frac{dx}{(x^2+1)^2} \\ &= 2\pi \left[\frac{x}{x^2+1} + \arctan x\right]_0^2 = 2\pi \left(\frac{2}{5} + \arctan 2\right) \approx 9.47 \text{ m}^3. \end{aligned}$$

24)

Since $y^2 = \operatorname{arcosh} x$, the volume is

$$V = \pi \int_1^2 \operatorname{arcosh} x \, dx.$$

Integration by parts gives $\int \operatorname{arcosh} x \, dx = x \operatorname{arcosh} x - \sqrt{x^2-1}$, so

$$V = \pi \left(2 \operatorname{arcosh} 2 - \sqrt{3}\right) = \pi \left(2 \ln(2+\sqrt{3}) - \sqrt{3}\right) \text{ units}^3.$$

25a)

With $y = (k - x^2)^{-1/4}$ and $0 \le x \le k$, the volume is

$$V = \pi \int_0^k \frac{dx}{\sqrt{k - x^2}} = \pi \left[\arcsin\left(\frac{x}{\sqrt{k}} \right) \right]_0^k = \pi \arcsin \sqrt{k}.$$

25b)

For $0 \le x \le \frac{1}{2}$,

$$V = \pi \int_0^{1/2} \frac{dx}{\sqrt{k - x^2}} = \pi \arcsin\left(\frac{1}{2\sqrt{k}} \right).$$

26)

The straight-line section gives $y = \sqrt{2}(x + 1)$ for $-1 \le x \le 0$. The curved section is $y = (2 - x)/\sqrt{x^2 + 2}$ for $0 \le x \le 2$. Thus

$$\begin{aligned} V &= \pi \int_{-1}^0 2(x+1)^2 \, dx + \pi \int_0^2 \frac{(2-x)^2}{x^2+2} \, dx \\ &= \frac{2\pi}{3} + \pi \left(\frac{8}{3} - \ln 9 + \sqrt{2} \arctan \sqrt{2} \right) - \frac{2\pi}{3} \\ &= \pi \left(\frac{8}{3} - \ln 9 + \sqrt{2} \arctan \sqrt{2} \right) \approx 5.72 \text{ units}^3. \end{aligned}$$

27a)

Using the shell method about the x-axis,

$$V = \pi \int_0^{7/2} (2y+1)^{2/3} \, dy = \frac{3\pi}{10} \left[(2y+1)^{5/3} \right]_0^{7/2} = \frac{93\pi}{10} \text{ units}^3.$$

27b)

$$V = \pi \int_{\pi/6}^{\pi/3} \operatorname{cosec}^2 y \, dy = \pi [-\cot y]_{\pi/6}^{\pi/3} = \frac{2\sqrt{3}\pi}{3} \text{ units}^3.$$

27c)

$$\begin{aligned} V &= \pi \int_0^{\pi/2} \frac{2 \sin y}{1 + \cos y} \, dy = -2\pi [\ln(1 + \cos y)]_0^{\pi/2} \\ &= \pi \ln 4 \text{ units}^3. \end{aligned}$$

27d)

$$\begin{aligned} V &= \pi \int_0^{\pi/4} y \sec^2 y \, dy \\ &= \pi [y \tan y + \ln |\cos y|]_0^{\pi/4} = \frac{\pi}{4}(\pi - \ln 4) \text{ units}^3. \end{aligned}$$

27e)

$$V = \pi \int_0^{\pi/4} (\sec y - \tan y)^2 \, dy$$
$$= \pi[2\tan y - 2\sec y - y]_0^{\pi/4} = \pi\left(4 - 2\sqrt{2} - \frac{\pi}{4}\right) \text{ units}^3.$$

28a)

The volume of one bead is found by rotating the semicircle about its diameter:

$$V = \pi \int_{-1.570}^{1.570} \left(\frac{5}{9} - \frac{2}{9}y^2 + \frac{1}{9}\cos\frac{\pi y}{3}\right) dy \approx 4.34 \text{ cm}^3.$$

28b)

The volume of 50 beads is approximately $50(4.34) = 217$ cm^3. This exceeds 215 cm^3, so there is enough material.

29)

Rotating $x = e^y$ about the y-axis gives

$$V = \pi \int_0^{\ln 4} e^{2y} \, dy = \frac{\pi}{2}\left[e^{2y}\right]_0^{\ln 4} = \frac{15\pi}{2} \text{ units}^3.$$

30)

The volume is

$$V = \pi \int_0^{\pi h/2} \left(1 - \sin\frac{y}{h}\right) dy = \pi\left[y + h\cos\frac{y}{h}\right]_0^{\pi h/2} = \frac{\pi^2 h}{2} - \pi h.$$

Given $V = 2\pi^2 - 4\pi$, this gives $h = 4$ m.

31a)

Using the volume of revolution and the given condition gives $k = \frac{3\pi}{2}$.

31b)

At the base, the radius is 2 cm, so the diameter is 4 cm.

31c)

Adding the cylindrical and curved sections gives a total volume of approximately

$$4.3356 + 152.7572 \approx 157 \text{ cm}^3.$$

31d)

The calculation assumes that the vessel walls have zero thickness. A real vessel has walls of finite thickness, so its internal capacity is slightly less than the calculated volume.

32a)

Using the condition on the top diameter gives $k = 6$.

32b)

The bowl's height is 6 cm.

32c)

At height h, the radius is $6 + 2\sin(\pi h/12)$. Hence

$$\begin{aligned} V(h) &= \pi \int_0^h \left(6 + 2\sin\frac{\pi y}{12}\right)^2 dy \\ &= 38\pi h - 288\cos\frac{\pi h}{12} - 12\sin\frac{\pi h}{6} + 288 \text{ cm}^3. \end{aligned}$$

32d)

The water surface radius at $h = 4$ is $6 + 2\sin(\pi/3)$. Since $dV/dt = 500$,

$$\frac{dh}{dt} = \frac{500}{\pi(6 + 2\sin(\pi/3))^2} \approx 2.66 \text{ cm min}^{-1}.$$

33a)

The volume of the coating is

$$V = \pi\left(\frac{e^{0.1}}{20} - \frac{11}{200}\right) \text{ m}^3 \approx 0.000812 \text{ m}^3.$$

33b)

Multiplying the volume by the density gives a mass of approximately 3.65 kg.

33c)

This is impractical because the coating would make the object too heavy; the model or the assumed density/thickness would need reconsideration.

34)

Using $x = \sinh y$, so that $x^2 = \sinh^2 y$, gives

$$\begin{aligned} V &= 2\pi \int_0^{\sinh 1} \operatorname{arsinh} x \, dx \\ &= 2\pi \left[x \operatorname{arsinh} x - \sqrt{x^2+1}\right]_0^{\sinh 1} = 2\pi\left(1 - \frac{1}{e}\right). \end{aligned}$$

35)

For C_1, rearrange to obtain $x^2 = \frac{k^2}{k-10}(y-10)$ for $k \le y \le 10$. For C_2,

$$x^2 = k\sqrt{k^2 - (y-k)^2}, \qquad 0 \le y \le k.$$

The volume is the sum of the volumes from the two intervals:

$$\begin{aligned} V &= \pi \int_k^{10} \frac{k^2}{k-10}(y-10)\,dy + \pi k \int_0^k \sqrt{k^2-(y-k)^2}\,dy \\ &= -\frac{1}{2}k^2(k-10)\pi + \frac{1}{4}k^3\pi^2 \\ &= \frac{1}{4}k^2\pi\big(k\pi - 2(k-10)\big) \text{ units}^3. \end{aligned}$$

36)

Rearranging the curve equation gives $x^2 = 1/(y^2+1)$. Therefore

$$V = \pi \int_0^1 \frac{dy}{1+y^2} = \pi[\arctan y]_0^1 = \frac{\pi^2}{4} \text{ units}^3.$$

37)

$$V = \pi \int_0^1 \frac{dy}{\sqrt{k-y^2}} = \pi \arcsin\left(\frac{1}{\sqrt{k}}\right).$$

38)

Using the given volume and the result from the previous question gives $k = 9$.

39a)

For $y = \operatorname{arcosh} x$, the inverse relation is $x = \cosh y$.

39b)

At $y = \frac{1}{2}\ln 5$, $x = \cosh y = \frac{1}{2}(\sqrt{5} + 1/\sqrt{5}) = 3/\sqrt{5}$. Hence

$$P = \left(\frac{3\sqrt{5}}{5}, \frac{1}{2}\ln 5\right).$$

39c)

The volume is

$$\begin{aligned} V &= \pi \int_0^{\frac{1}{2}\ln 5} \cosh^2 y\,dy = \pi\left[\frac{y}{2} + \frac{\sinh 2y}{4}\right]_0^{\frac{1}{2}\ln 5} \\ &= \pi\left(\frac{1}{4}\ln 5 + \frac{3}{5}\right) \text{ units}^3. \end{aligned}$$

40a)

The horizontal asymptote gives $a = 4$. The vertical asymptote gives $b = 3$.

40b)

With $a = 4$ and $b = 3$, the curve is $y = (4x+7)/(x+3)$.

40c)

$$\begin{aligned} V &= \pi \int_{-7/4}^{0} \left(4 - \frac{5}{x+3}\right)^2 dx \\ &= \pi \left[16x - 40\ln|x+3| - \frac{25}{x+3}\right]_{-7/4}^{0} \\ &= \pi \left(\frac{119}{3} - 40\ln\frac{12}{5}\right) \text{ units}^3. \end{aligned}$$

40d)

$$\begin{aligned} V &= \pi \int_{0}^{7/3} \left(\frac{7-3y}{y-4}\right)^2 dy \\ &= \pi \int_{0}^{7/3} \left(-3 - \frac{5}{y-4}\right)^2 dy \\ &= \pi \left[9y + 30\ln|y-4| - \frac{25}{y-4}\right]_{0}^{7/3} \\ &= \pi \left(\frac{119}{4} - 30\ln\frac{12}{5}\right) \text{ units}^3. \end{aligned}$$

41a)

$$V = \pi \int_0^1 \sin^2\left(\frac{\pi x}{2}\right) dx = \frac{\pi}{2} \int_0^1 (1 - \cos \pi x) dx = \frac{\pi}{2} \text{ units}^3.$$

41b)

Using $x = \frac{2}{\pi} \arcsin y$,

$$\begin{aligned} V &= \frac{4}{\pi} \int_0^1 \arcsin^2 y \, dy \\ &= \frac{4}{\pi} \left[y \arcsin^2 y + 2 \arcsin y \sqrt{1-y^2} - 2y\right]_0^1 \\ &= \pi - \frac{8}{\pi} \text{ units}^3. \end{aligned}$$